

МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ
РОССИЙСКОЙ ФЕДЕРАЦИИ
ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ БЮДЖЕТНОЕ ОБРАЗОВАТЕЛЬНОЕ
УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ
«БЕЛГОРОДСКИЙ ГОСУДАРСТВЕННЫЙ
ТЕХНОЛОГИЧЕСКИЙ УНИВЕРСИТЕТ им. В. Г. ШУХОВА»
(БГТУ им. В.Г. Шухова)

Кафедра программного обеспечения вычислительной техники и автоматизированных
систем

Лабораторная работа №2
по дисциплине «Теория надежности»
тема: «Игра в имитацию»

Выполнил: ст. группы ВТ-32
Воскобойников И. С.
Проверил: Кабелянц П. С.

Белгород 2021 г.

Задание:

Дискретная случайная величина ξ принимает значения 0, 1, 2 и 3 с вероятностями соответствующими данным вариантам. Используя генератор псевдослучайных чисел, получить 10000 чисел, имитирующих поведение ξ . Проверить на уровне значимости 0,05 соответствие полученных данных распределению ξ , используя критерий сравнения долей. Вероятности: $p_0=i/2*(i+j+k)$, $p_1=(j+k)/2*(i+j+k)$, $p_2=(i+k)/2*(i+j+k)$, $p_3=j/2*(i+j+k)$. Значения i, j, k являются последними тремя цифрами зачетки. Если какая-то из этих трех цифр равна 0, то вместо 0 берется 10.

Результат работы программы:

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File - main

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File - main

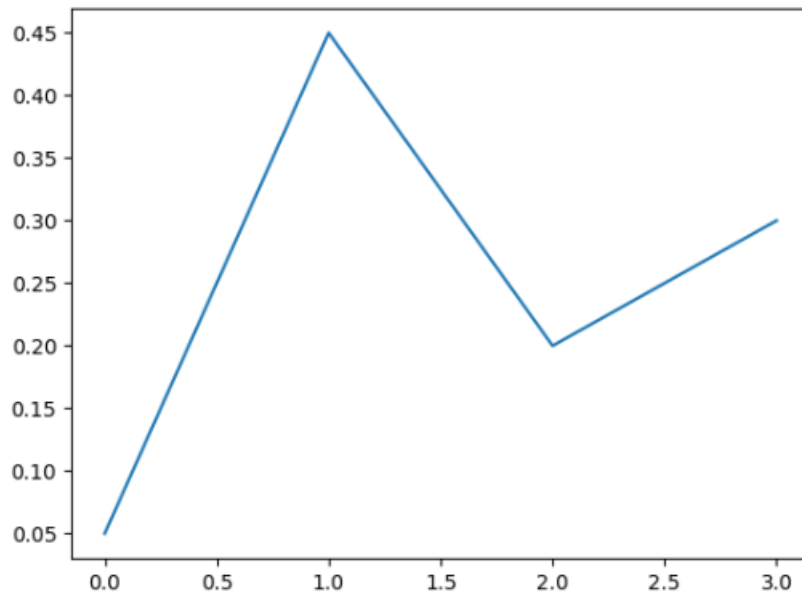
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Код программы:

```

from random import random
from scipy.stats import chisquare
from math import sqrt
from matplotlib import pyplot as plt

# Последние 3 цифры зачетки
i = 1
j = 10
k = 10

# Вероятности
p_list = [
    i/(2*(i+j+k)),
    (j+k)/(2*(i+j+k)),
    (i+k)/(2*(i+j+k)),
    j/(2*(i+j+k))
]
print("Вероятности")
for i, p in enumerate(p_list):
    print(f"P{i} = {p}")

class Discrete_Random_Variable:
    def __init__(self, p_list):
        self.p_list = p_list
        self.split_segment = self.get_split_segment(p_list)

    @staticmethod
    def get_split_segment(p_list):
        split_segment = p_list.copy()
        for i in range(len(split_segment)-1):
            split_segment[i+1] += split_segment[i]
        return split_segment

    def get_drv_value(self):
        random_value = random()
        segment = 0

```

```

        while random_value >= self.split_segment[segment]:
            segment += 1

    return segment

drv = Discrete_Random_Variable(p_list)
n = 100000
generated_p_values = [drv.get_drv_value() for _ in range(n)]
print("Ваша выборка (100000)")
print(generated_p_values)

p = [current_p*n for current_p in p_list]
print("Теоретические частоты")
print(p)

generated_p = [generated_p_values.count(p) for p in range(len(p_list))]
print("Наблюдаемые частоты")
print(generated_p)

K, P = chisquare(generated_p, p)
print(f"Хи^2 K={K}, p={P}\n")
print("Сравнение долей")
for p_index in range(len(p)):
    gen_p = generated_p[p_index] / n
    teor_p = p[p_index] / n
    print(f"Вероятность из полученной выборки {gen_p}\n"
          f"Теоретическая вероятность {teor_p}")
    print(f"K для P{i} = {abs((gen_p - teor_p) / (sqrt(gen_p * (1-gen_p) /
n)))}\n")

print("Выборка")
plt.plot(generated_p)
plt.show()

print("Заданное распределение")
plt.plot(p_list)
plt.show()

```